Non-stationary models II

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ECMA 31150, Spring 2026

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Plan for this slide deck

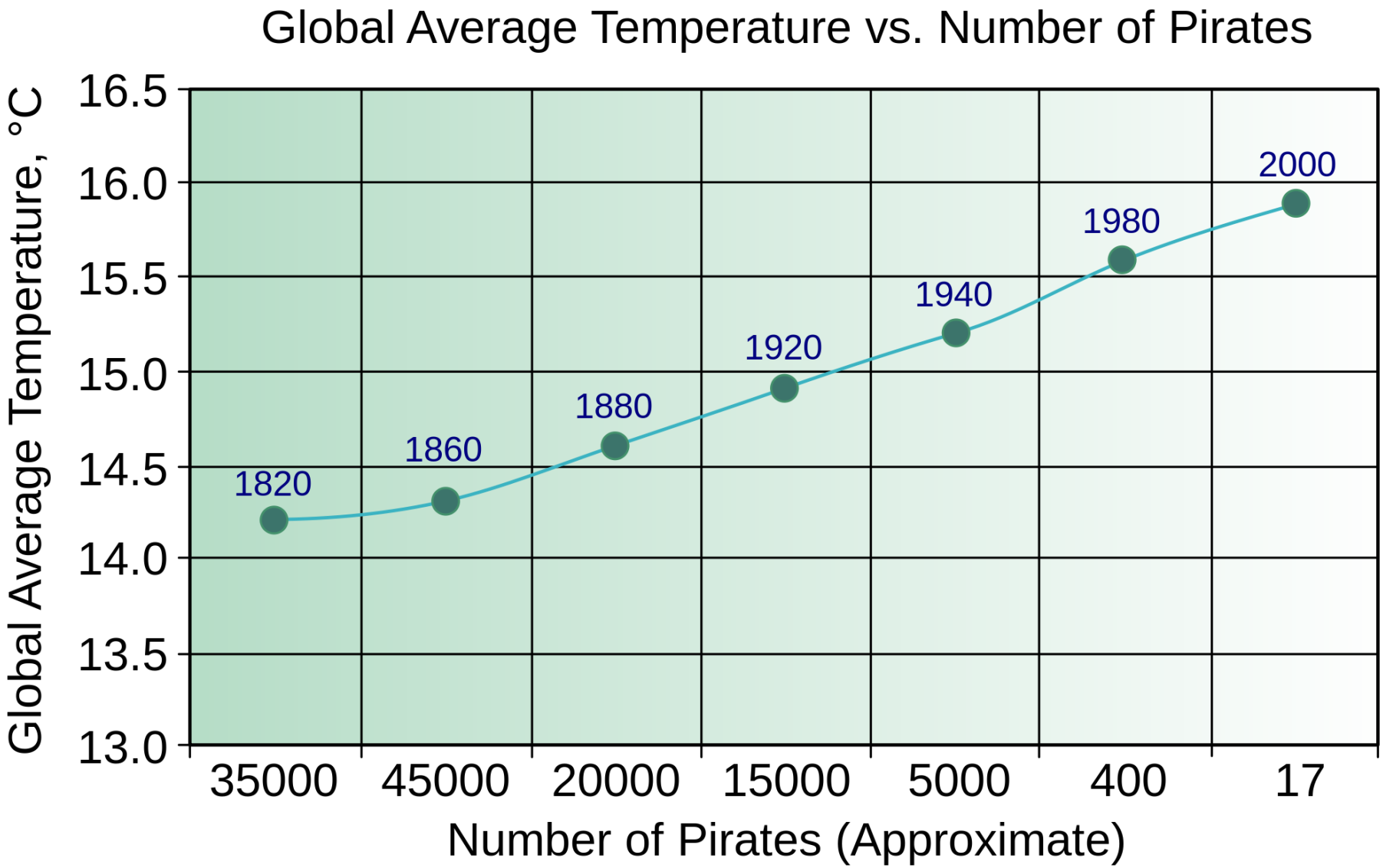
1. Regressions involving two trending variables
2. Modeling multiple time series with common trends
3. Testing for structural breaks
4. Removing trends and seasonality from data

Spurious regression

- Assume the two series $\{y_t\}$ and $\{x_t\}$ are driftless random walks that are completely independent of each other.
- Suppose we nevertheless run an OLS regression (without lags):

$$y_t = \hat{c} + \hat{\beta} x_t + \hat{v}_t.$$

- Then $\hat{\beta}$ does not converge to 0, and the magnitude of the t-statistic for testing $\beta = 0$ diverges to ∞ as $T \to \infty$!
  - The R^2 value converges to a non-degenerate random variable in $[0, 1]$.
- **Spurious regression:** a regression of two independent trending series on each other can spuriously indicate a significant relationship if we interpret the OLS output naively.
  - Intuitively, the stochastic trends in the series act as "omitted variables".



Source: `https://commons.wikimedia.org/wiki/File:PiratesVsTemp(en).svg`

Common stochastic trends

- Since regressions using variables with *independent* trends can be spurious, we need a framework for evaluating whether trends are independent or shared.
- Let's consider the case where the two I(1) series y_t and x_t share the *same* stochastic trend:

$$y_t = \lambda_1 w_t + v_{yt},$$
$$x_t = \lambda_2 w_t + v_{xt},$$

where w_t is an I(1) series and (v_{yt}, v_{xt}) are covariance stationary.

Cointegration

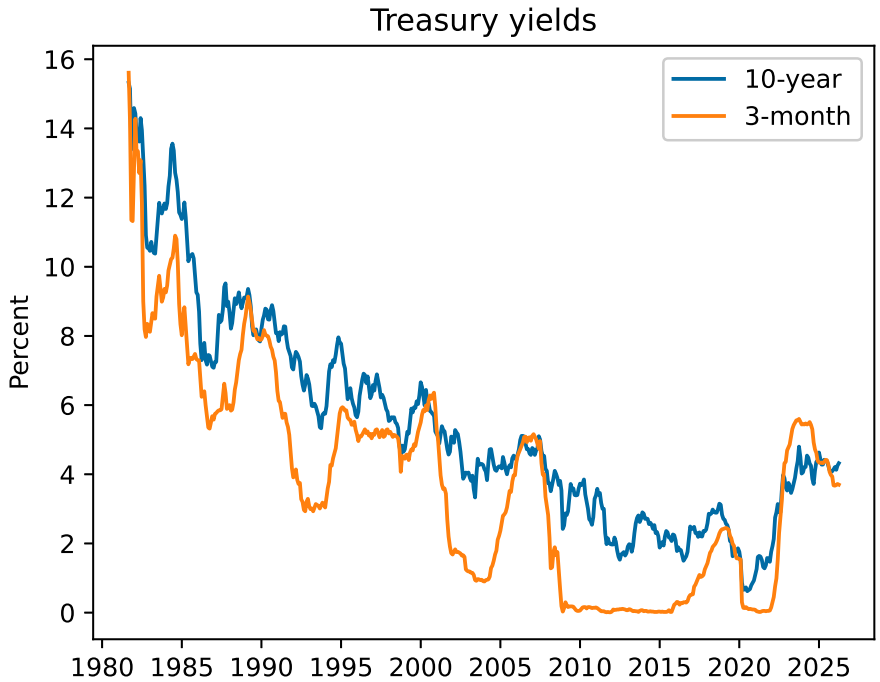
$$y_t = \lambda_1 w_t + v_{yt},$$
$$x_t = \lambda_2 w_t + v_{xt},$$

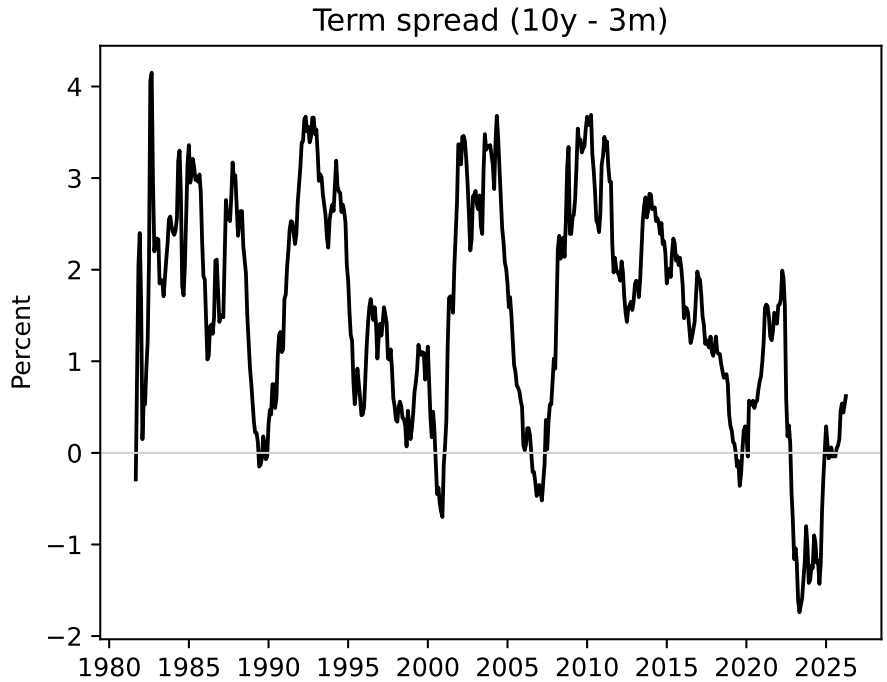
- Though each of y_t and x_t is non-stationary, a *linear combination* of them is covariance stationary:

$$y_t - \frac{\lambda_1}{\lambda_2} x_t = \underbrace{\left(\lambda_1 - \frac{\lambda_1}{\lambda_2}\lambda_2\right)}_{=0} w_t + \underbrace{v_{yt} - \frac{\lambda_1}{\lambda_2} v_{xt}}_{\text{cov. stationary}}.$$

  (Any scalar multiple of this linear combination also works.)

- (y_t, x_t) are **cointegrated**: they have a stable long-run relationship.





The spread $y_{10y,t} - y_{3m,t}$ looks roughly stationary, even though each individual yield is trending

Testing for a known cointegrating relationship

- Consider two trending series y_t and x_t. They may share a common stochastic trend (cointegration) or each follow their own trend.
- Suppose we want to test for cointegration with a *known* cointegrating relationship $y_t - \beta x_t$.
  - E.g., $\beta = 1$ for the yield spread on the previous slide.
- First we run two ADF tests to test whether each of y_t and x_t is individually I(1).
  - If we reject the null of a unit root for either, then they obviously can't share a common stochastic trend.
- Then we run another ADF test on the purported cointegrating relationship $y_t - \beta x_t$. If we reject a unit root, then we reject the null hypothesis that the series are *not* cointegrated.

```
# ADF unit root test for each series (constant + trend)
for s in ['GS3M', 'GS10']:
    stat, pval, nlags, *_ = adfuller(df[s], regression='ct', autolag='AIC')
    print(f'{s}: ADF stat = {stat:.3f}, p-value = {pval:.3f}, lags = {nlags}')

# Test known cointegrating relationship (no trend)
stat, pval, nlags, *_ = adfuller(df['spread'], regression='c', autolag='AIC')
print(f'Spread: ADF stat = {stat:.3f}, p-value = {pval:.3f}, lags = {nlags}')
```

Results:

```
GS3M: ADF stat = -2.856, p-value = 0.177, lags = 10
GS10: ADF stat = -2.971, p-value = 0.140, lags = 3
Spread: ADF stat = -3.874, p-value = 0.002, lags = 9
```

We fail to reject the null that each series is individually I(1). We do reject the null of no cointegration. (5% significance level.)

Estimating cointegrating relationships

- If two series y_t and x_t are cointegrated, we can estimate the cointegrating relationship consistently by running an OLS regression:

$$y_t = \hat{c} + \hat{\beta} x_t + \hat{v}_t.$$

  The estimated cointegrating relationship is $y_t - \hat{\beta} x_t$.

- $\hat{\beta}$ is consistent under cointegration (in fact, super-consistent).

- However, because of the stochastic trend, the OLS estimator $\hat{\beta}$ is non-normally distributed, so we cannot rely on the usual t-test or confidence interval.

  - There exist other estimation approaches that have a normal limiting distribution. See Stock & Watson textbook, chapter 17.4.

OLS Regression Results

Dep. Variable:	GS10	R-squared:	0.845
Model:	OLS	Adj. R-squared:	0.845
Method:	Least Squares	F-statistic:	2914.
Date:	Wed, 20 May 2026	Prob (F-statistic):	2.05e-218
Time:	15:30:39	Log-Likelihood:	-859.91
No. Observations:	536	AIC:	1724.
Df Residuals:	534	BIC:	1732.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	1.9185	0.082	23.267	0.000	1.757	2.081
GS3M	0.9007	0.017	53.977	0.000	0.868	0.933

Recall: t-test and conf. interval are invalid, as they rely on normality

Testing for an unknown cointegrating relationship

- To test whether y_t and x_t are cointegrated with an *unknown* cointegrating relationship:
  1. Estimate the cointegrating relationship by an OLS regression of y_t on x_t (as on previous slide).
  2. Run an ADF test on $y_t - \hat{\beta} x_t$. If it rejects a unit root, then we reject the null of no cointegration.
- However, it turns out that the estimation error in $\hat{\beta}$ changes the limiting distribution of the ADF t-statistic. So the critical values for the second step must be adjusted relative to the case with known β.
- This **Engle & Granger ADF** (EG-ADF) test with corrected critical values is built into many software packages.

```
# EG-ADF test with estimated cointegrating relationship
stat, pval, _ = coint(df['GS10'], df['GS3M'], trend='c', autolag='AIC')
print(f'EG-ADF stat = {stat:.3f}, p-value = {pval:.3f}')
```

Results:

```
EG-ADF stat = -3.450, p-value = 0.037
```

We reject the null of no cointegration at the 5% level.

Error correction model

- If y_t and x_t are cointegrated, then changes in these series should be predictable by the cointegrating relationship:

$$\Delta y_t = c_y + A_{yy}\Delta y_{t-1} + A_{yx}\Delta x_{t-1} + A_{yc}(y_{t-1} - \beta x_{t-1}) + u_{yt},$$
$$\Delta x_t = c_x + A_{xy}\Delta y_{t-1} + A_{xx}\Delta x_{t-1} + A_{xc}(y_{t-1} - \beta x_{t-1}) + u_{xt}.$$

- **Error correction model**: if y_t is currently far away from βx_t, then we expect these to move closer together in the future, because their difference is covariance stationary.

- If we naively run a VAR in the first differences $(\Delta y_t, \Delta x_t)$, we lose out on the predictive power of the cointegrating relationship (a.k.a. error correction term).

- Running a VAR in the levels (y_t, x_t) is more robust as it allows for cointegrating relationships among the variables.

Regression with trending variables: summary

- We should be careful with regressions involving trending variables.
- If we have theoretical reasons to expect cointegration (shared trend), we could test for it or impose a known cointegrating relationship directly in an error correction model.
- For many purposes it suffices to just run a VAR (or LP) with a generous number of lags. The lags solve the spurious regression problem.
- The cointegration concept can be extended to relationships involving more than two variables. See Stock & Watson textbook, chapter 17.4.

Structural breaks

- **Structural breaks**: changes in the data generating process over time. Different kind of non-stationarity from trends.
- We've already discussed how to use state space models to analyze time-variation in parameters that is (a) *gradual and period-by-period* or (b) *discrete and intermittent*.
- In some applications, we might instead expect that the parameters have undergone a *large, one-time* break.
- Let's consider the regression example:

$$y_t = c + \beta_t x_t + v_t, \quad \beta_t = \begin{cases} \beta & \text{for } t < t^*, \\ \beta + \delta & \text{for } t \geq t^*, \end{cases}$$

where t^* is the break date and $\delta \neq 0$ is the break magnitude.

Testing for break at a known date

- If the potential break date t^* is *known* to us, we can test for breaks using a **Chow test**:

  1. Define the break indicator

$$d_t \equiv \begin{cases} 0 & \text{for } t < t^*, \\ 1 & \text{for } t \geq t^*. \end{cases}$$

  2. Run an OLS regression with interaction term:

$$y_t = \hat{c} + \hat{\beta} x_t + \hat{\delta}(x_t \times d_t) + \hat{v}_t.$$

  3. Test the null hypothesis $\delta = 0$ (no break) with a t-test.

- More generally, if we want to test for breaks in coefficients on multiple covariates, we include interactions of d_t with each of those. Then do an F-test for the joint significance of the coefficients on the interaction terms.

```
# Load data
gdp = pd.read_csv('data/GDPC1.csv', index_col='observation_date',
  parse_dates=True)
spread = pd.read_csv('data/T10Y3MM.csv', index_col='observation_date',
  parse_dates=True)

gdp['gdp_growth'] = 400 * np.log(gdp['GDPC1']).diff()
spread_q = spread.resample('QS').mean()

df_adl = gdp[['gdp_growth']].join(spread_q, how='inner')
df_adl = df_adl.loc[:'2019'] # pre-2020 sample

# Full-sample ADL regression
df_adl['gdp_growth_lag1'] = df_adl['gdp_growth'].shift(1)
df_adl['gdp_growth_lag2'] = df_adl['gdp_growth'].shift(2)
df_adl['spread_lag1']     = df_adl['T10Y3MM'].shift(1)
df_adl['spread_lag2']     = df_adl['T10Y3MM'].shift(2)

covariates = ['gdp_growth_lag1', 'gdp_growth_lag2', 'spread_lag1', 'spread_lag2']
X_adl = sm.add_constant(df_adl[covariates])
res_adl = sm.OLS(df_adl['gdp_growth'], X_adl, missing='drop').fit(cov_type='HC1')
print(res_adl.summary())
```

OLS Regression Results

Dep. Variable:	gdp_growth	R-squared:	0.260
Model:	OLS	Adj. R-squared:	0.239
Method:	Least Squares	F-statistic:	11.49
Date:	Wed, 20 May 2026	Prob (F-statistic):	3.94e-08
Time:	15:30:39	Log-Likelihood:	-322.65
No. Observations:	150	AIC:	655.3
Df Residuals:	145	BIC:	670.4
Df Model:	4		
Covariance Type:	HC1		

	coef	std err	z	P>|z|	[0.025	0.975]
const	0.9231	0.409	2.259	0.024	0.122	1.724
gdp_growth_lag1	0.3438	0.105	3.280	0.001	0.138	0.549
gdp_growth_lag2	0.2128	0.080	2.652	0.008	0.056	0.370
spread_lag1	-0.2585	0.304	-0.849	0.396	-0.855	0.338
spread_lag2	0.4460	0.329	1.355	0.175	-0.199	1.091

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

```
def chow_test(break_date):
    brk = (df_adl.index >= break_date).astype(float)
    breaks = pd.DataFrame({
        'break_spread_lag1': brk * df_adl['spread_lag1'],
        'break_spread_lag2': brk * df_adl['spread_lag2'],
    })
    X = sm.add_constant(pd.concat([df_adl[covariates], breaks], axis=1))
    res = sm.OLS(df_adl['gdp_growth'], X, missing='drop').fit(cov_type='HC1')
    f_test = res.f_test('break_spread_lag1 = break_spread_lag2 = 0')
    return res, f_test

# Chow test for 2008q3
res_chow, f_chow = chow_test('2008-07-01')
print(res_chow.summary())
print('\nChow test')
print(f_chow)
```

Note: here we're only testing for breaks in the coefficients on the lagged spread (not lagged GDP growth or the intercept)

OLS Regression Results

Dep. Variable:	gdp_growth	R-squared:	0.286
Model:	OLS	Adj. R-squared:	0.256
Method:	Least Squares	F-statistic:	8.387
Date:	Wed, 20 May 2026	Prob (F-statistic):	8.47e-08
Time:	15:30:39	Log-Likelihood:	-319.96
No. Observations:	150	AIC:	653.9
Df Residuals:	143	BIC:	675.0
Df Model:	6		
Covariance Type:	HC1		

	coef	std err	z	P>|z|	[0.025	0.975]
const	1.1067	0.420	2.634	0.008	0.283	1.930
gdp_growth_lag1	0.3002	0.099	3.027	0.002	0.106	0.494
gdp_growth_lag2	0.1818	0.087	2.096	0.036	0.012	0.352
spread_lag1	-0.1262	0.334	-0.378	0.706	-0.782	0.529
spread_lag2	0.4610	0.364	1.266	0.206	-0.253	1.175
break_spread_lag1	-1.0399	0.939	-1.108	0.268	-2.880	0.800
break_spread_lag2	0.6386	0.892	0.716	0.474	-1.111	2.388

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

Chow test
<F test: F=2.209285321811282, p=0.11351275499888673, df_denom=143, df_num=2>

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Notes:
[1] Standard Errors are heteroscedastic

Fail to reject
null of no break

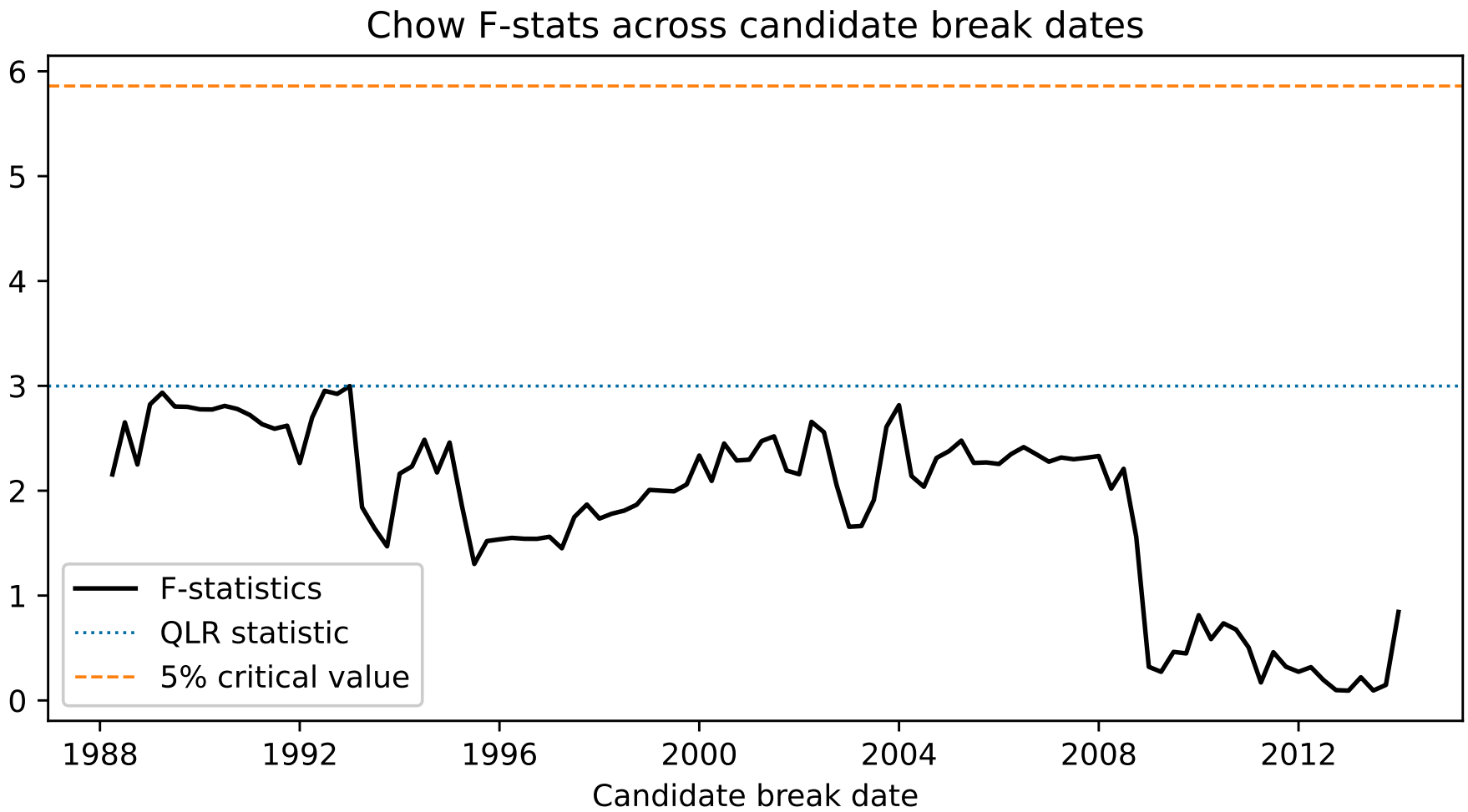
Chow test
<F test: F=2.209285321811282, p=0.11351275499888673, df_denom=143, df_num=2>

Testing for break at an unknown date

- Often we don't know the exact break date t^* *a priori*.
- Instead, we can run a Chow test for *every possible* candidate break date t^* in some range and record the **largest** F-statistic. This is called the **Quandt likelihood ratio** (QLR) statistic.
  - To avoid small-sample issues near the boundaries, it's common to restrict the candidate break dates to the central 70% of the sample.
- We can't compare the QLR-stat with the usual critical value for an F-test because of the **multiple testing** problem: we're implicitly testing for the null of no break at very many time points. The probability of type-I error compounds across tests.
- Critical values for the QLR test that adjust for multiple testing can be found in Stock & Watson, chapter 15.8.

```
# QLR test
sample = X_adl.dropna().index
trim = 0.15
i_lo = int(np.ceil(trim * len(sample)))
i_hi = int(np.floor((1 - trim) * len(sample)))
break_dates = sample[i_lo:i_hi] # central 70% of sample

f_stats = pd.Series(index=break_dates, dtype=float)
for t in break_dates:
    _, f_test = chow_test(t)
    f_stats[t] = f_test.statistic
```



QLR critical value = 5.86 (for 2 restrictions, 15% sample trimming at each end, 5% significance level)

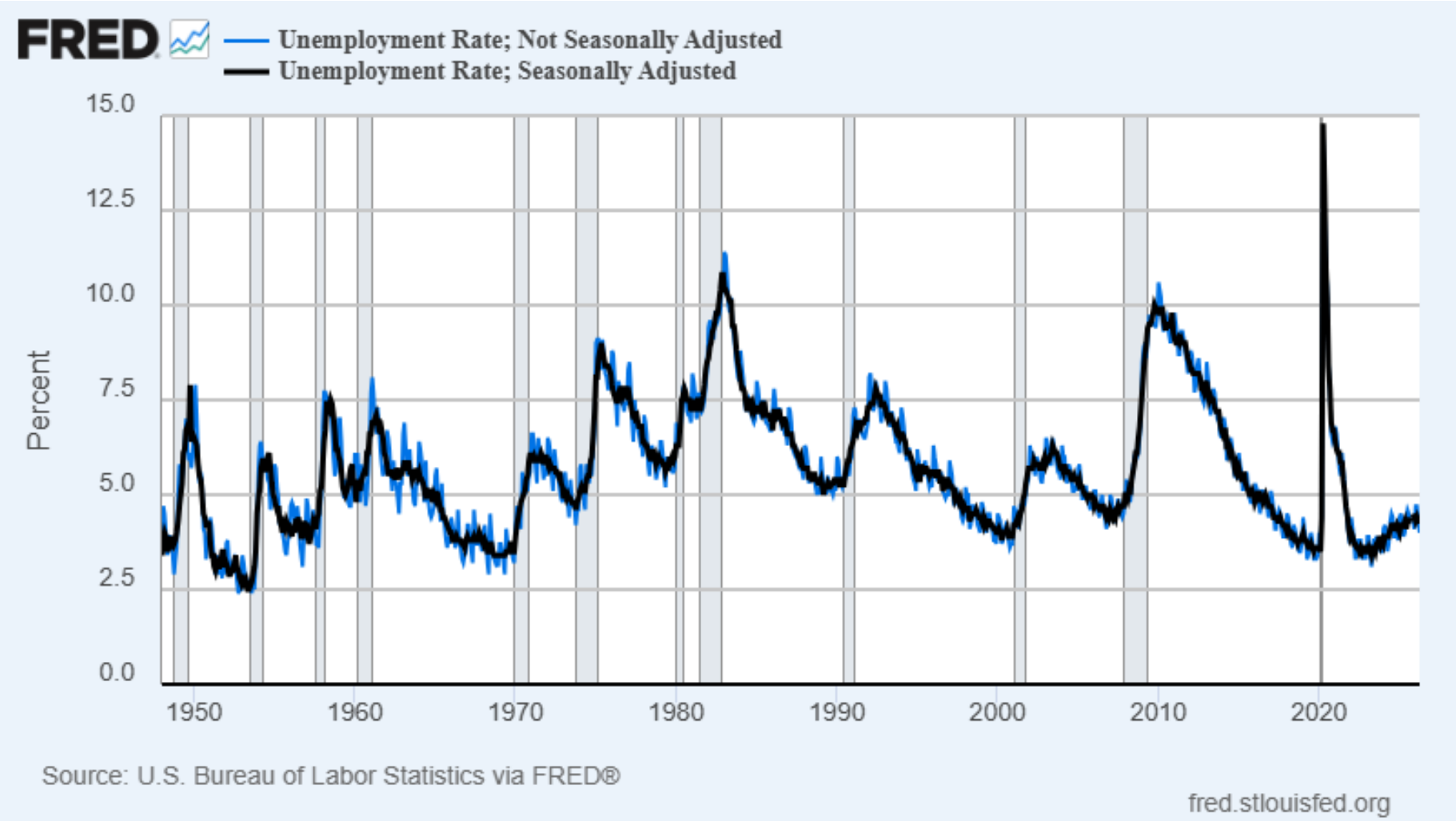
Here we fail to reject the null hypothesis of no breaks

Estimated break date, multiple breaks

- If the QLR test detects a break, the time point t^* yielding the largest F-stat is an estimator of the break date.
  - For subsequent analysis, we can estimate our model separately on each subsample $t \in [1, t^* - 1]$ and $t \in [t^*, T]$.
- If we suspect multiple large breaks, we can apply the QLR test again on the subsamples.
- The QLR test usually also has decent power against the alternative hypothesis of gradual time-variation (as opposed to one large break), though in this case the estimated break date is meaningless.

Seasonality

- Some time series exhibit **seasonality**: they behave systematically differently at some points of the year than at others.
  - For example, due to holidays or annual temperature cycles.
- This is yet another form of non-stationarity.
- We can try to control for seasonal effects by including dummy variables for each within-year "season" in our regressions.
  - E.g., in quarterly data, control for 3 dummy variables for 1st, 2nd, and 3rd quarter of the year.
- But usually best to just use **seasonally adjusted** data supplied by statistical agencies. They use more advanced techniques to remove seasonal patterns.


FRED
Unemployment Rate; Not Seasonally Adjusted
Unemployment Rate; Seasonally Adjusted
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Percent
1950
1960
1970
1980
1990
2000
2010
2020
Source: U.S. Bureau of Labor Statistics via FRED®
fred.stlouisfed.org

Detrending

- When working with trending data, some papers choose to **detrend** their data prior to analysis.
- The most common procedure is the **Hodrick-Prescott filter**, which fits a slow-moving trend to the data and then subtracts that estimated trend.
- But the inevitable estimation error in the trend can impart spurious dynamics in the "detrended" residual.
  - See Hamilton (2018) paper on the syllabus for discussion.
- Detrending also removes any predictive power arising from cointegrating relationships.
- Usually best to avoid detrending if possible and instead tackle stochastic trends directly, as we've discussed.

What did we learn?

- Spurious regression
- Cointegration
  - Known cointegrating relationship: ADF test
  - Unknown cointegrating relationship: EG-ADF test
  - Error correction model, naive differencing loses information
- Testing for structural breaks
  - Known break date: Chow test
  - Unknown break date: QLR test
- Seasonality, detrending